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Audio Emotion Recognition

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1. LITERATURE REVIEW

Speech Emotion Recognition (SER) focuses on automatically identifying human emotions from speech signals and is widely used in human-computer interaction, healthcare monitoring, call center analytics, and conversational agents. A large body of work has explored both handcrafted feature pipelines and end-to-end neural approaches.

1.1 CLASSICAL SER: HANDCRAFTED FEATURES + SHALLOW MODELS

Earlier SER systems typically used handcrafted acoustic features such as Mel-Frequency Cepstral Coefficients (MFCCs), pitch, energy, formants, jitter, shimmer, and spectral descriptors. These features were summarized over time (mean, variance, percentiles) and fed into classifiers like Support Vector Machines (SVM), Gaussian Mixture Models (GMM), or Hidden Markov Models (HMM). Studies show MFCC + SVM pipelines achieving around **60-70%** accuracy on standard datasets such as EMO-DB and RAVDESS for 6-7 emotion classes.

These methods are simple and interpretable but depend heavily on feature engineering and often struggle with cross-corpus generalization and subtle emotional differences.

1.2 DEEP LEARNING: SPECTROGRAMS, CNNs AND RNNs

With deep learning, SER models increasingly use **time-frequency representations** (e.g., log-Mel spectrograms) and apply CNNs, LSTMs, or hybrid CNN-LSTM models. These architectures can learn local temporal and spectral patterns from raw features and have been shown to outperform classical MFCC + SVM approaches, especially when enough labeled data is available.

1.3 SELF-SUPERVISED SPEECH TRANSFORMERS (Wav2Vec2, HuBERT)

More recently, self-supervised speech transformers like **Wav2Vec2** and **HuBERT** have become the dominant approach. They are pre-trained on large unlabeled speech corpora and can be fine-tuned on relatively small SER datasets, achieving state-of-the-art performance.

Several works show that features from pre-trained Wav2Vec2, combined with light downstream networks, significantly improve SER accuracy and robustness compared to MFCC-based baselines.

1.4 DATASETS: RAVDESS, CREMA-D, TESS, SAVEE

The dataset used in this assignment is a merged corpus constructed from four established emotional speech datasets:

- **RAVDESS** – 24 actors producing emotional speech and song in North American English, with validated labels for anger, happiness, sadness, fear, disgust, surprise, calm, and neutral.
- **CREMA-D** – 7,442 clips from 91 actors expressing six emotions (Anger, Disgust, Fear, Happy, Neutral, Sad) with crowd-sourced ratings.
- **TESS** – Emotional speech set designed to study aging and emotion perception, with multiple emotions spoken by older/adult speakers.
- **SAVEE** – English audiovisual emotional database recorded from four male actors, widely used in SER research.

Merging these corpora introduces speaker, recording, and style variability, making the task closer to real-world conditions and more challenging than training on a single corpus.

2. APPROACHES STUDIED

In this assignment, three different SER frameworks were implemented and evaluated on the same train/validation/test split:

- MFCC + SVM (Classical Baseline)
- Zero-Shot Transformer Model (Pretrained)
- Fine-Tuned Wav2Vec2 Model (End-to-End)

Each approach represents a different “generation” of SER methodology.

2.1 MFCC + SVM BASELINE

Classical SER systems rely on handcrafted acoustic features such as MFCCs, pitch, energy, jitter, shimmer, and spectral descriptors. MFCCs are the most widely used because they approximate how humans perceive sound frequencies.

MFCC Processing Steps:

- **Pre-emphasis:** Applied to enhance high-frequency components that often reflect emotional intensity, such as those present in anger or fear.
- **Framing (20–25 ms frames):** Speech is quasi-stationary in short windows. Emotional cues like pitch changes happen across frames.
- **Hamming Windowing:** Reduces spectral leakage.
- **FFT:** Converts the signal into the frequency domain.

- Mel Filterbanks: Maps frequencies onto the perceptual Mel scale, emphasizing human hearing sensitivity.
- Log Compression: Captures loudness variations relevant to emotion (e.g, shouting vs sadness).
- Discrete Cosine Transform (DCT): Produces compact cepstral coefficients (MFCCs).

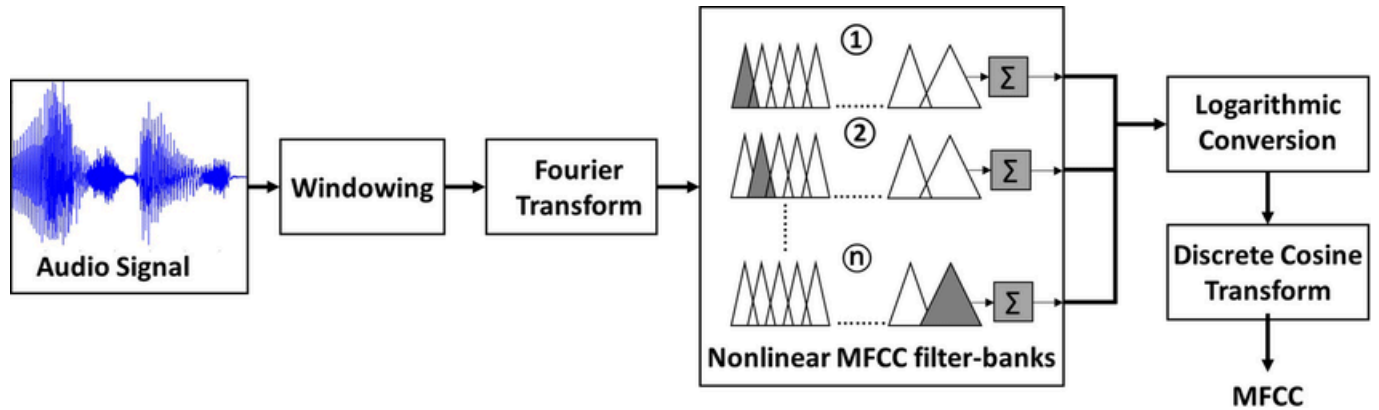


Figure 1: MFCC Feature Extraction Process (Source: Abeysinghe et al., 2021)

Feature Extraction:

- All audio files were resampled to 16 kHz.
- 40 MFCC coefficients were computed using Librosa.
- To obtain a fixed-length representation independent of utterance duration:
 - The mean and standard deviation of each MFCC coefficient were computed across time.
- Final feature vector = 80 dimensions (40 means + 40 standard deviations).
- Features standardized with StandardScaler

Reason for Mean and Standard Deviation:

These statistics provide a global summary of the spectral characteristics and capture variability within the signal. They create a compact, fixed-length representation suitable for classical models like SVM.

Classifier:

- SVM with RBF kernel ($C = 10$, $\gamma = \text{"scale"}$)
- Trained on training set MFCC features
- Evaluated on validation and test sets

An SVM with an RBF kernel handles non-linear class boundaries and works well with medium-sized feature vectors. This makes it a strong baseline for MFCC-based SER.

Expected Limitations:

- MFCCs cannot capture long-term prosody (rhythm, pauses).
- SVMs lack temporal modeling ability.
- Sensitive to dataset variability (different microphones, actors, corpora).

This pipeline is representative of traditional SER systems and directly comparable to MFCC-based literature.

2.2 ZERO-SHOT MODEL

The second approach evaluates whether a pretrained SER transformer can generalize to our merged dataset without any fine-tuning.

Model Details:

- Pretrained Wav2Vec2-based speech emotion model
HuggingFace model: ehcalabres/wav2vec2-lg-xlsr-en-speech-emotion-recognition
It includes:
 - Approximately 300M parameters
 - 24 transformer encoder layers
 - Self-supervised multilingual pretraining
 - A classification head trained on external SER corpora

Label Mapping:

The pretrained model outputs 8 emotions, including Calm, while our dataset uses 7 emotions.

- Calm was mapped to Neutral, as both represent low-arousal, low-activation emotional states.
- The remaining labels matched directly to our dataset's categories.

This mapping allows consistent evaluation without discarding valid predictions.

Pipeline:

- Each test audio file passed to the model
- Predicted emotion label returned directly
- Mapped to our 7-label set where possible

Expected Behaviour:

- Strong tendency to predict Neutral or other majority classes
- Poor recall for high-arousal emotions (Fearful, Disgusted, Surprised)
- Significantly lower performance compared to trained models

This approach measures how well a model trained on different emotion corpora generalizes to this merged dataset, in a strict zero-shot setting.

2.3 FINE-TUNED Wav2Vec2 MODEL

This is the most advanced approach, leveraging self-supervised representation learning.

Wav2Vec2 learns powerful speech features directly from raw audio, enabling high SER accuracy after fine-tuning.

Model:

- facebook/wav2vec2-base (self-supervised transformer encoder)
- Classification head adapted for 7 emotion classes
- Fine-tuned end-to-end on raw audio

Wav2vec 2.0 Pre-training

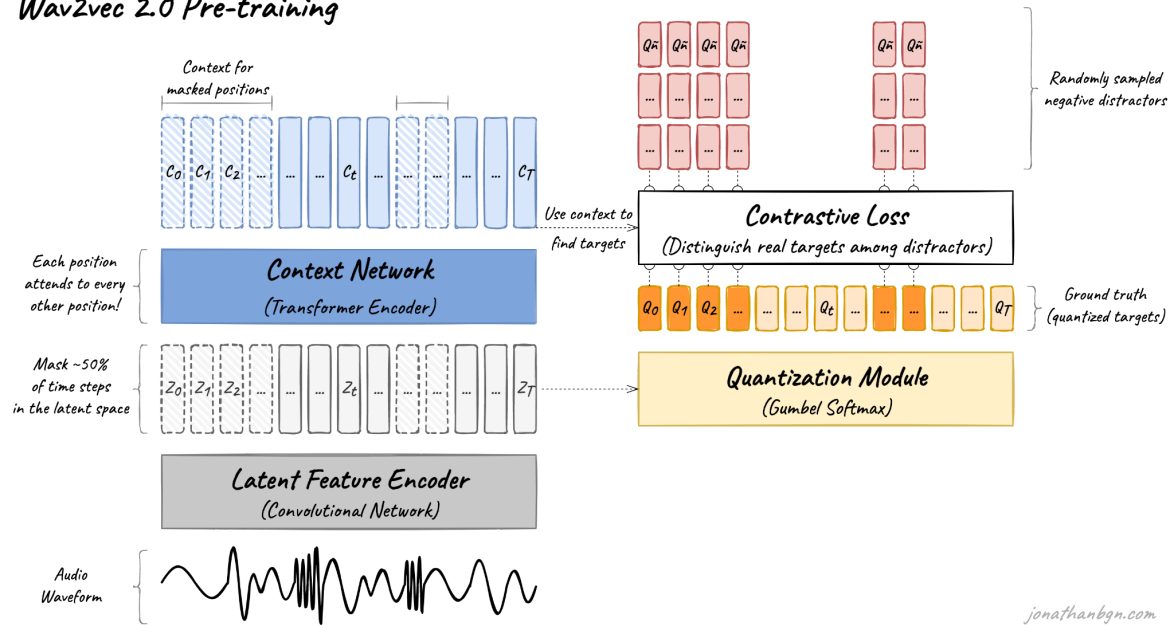


Figure 2: Wav2Vec2 Transformer Architecture (Source: Baevski et al., 2020)

Training Setup:

- Input: raw waveforms, resampled to 16 kHz
- Loss: Cross-entropy
- Epochs: 5 (with validation monitoring)
- Optimizer & schedule: controlled via HuggingFace TrainingArguments
- Batch size and gradient accumulation set to fit within Colab constraints

Training Pipeline:

- Raw waveforms passed directly to Wav2Vec2
- Transformer encoder learns contextual emotional patterns
- Classification head outputs 7-class predictions
- All parameters updated during training

Expected Advantages:

- Learns emotional cues directly from waveform (no handcrafted features)
- Captures long-range prosody and temporal dynamics
- More robust across speakers and datasets
- Highest overall accuracy among the three approaches

This approach represents a modern, end-to-end SER system consistent with recent research that fine-tunes Wav2Vec2 for emotion tasks.

3. HYPOTHESIS

Before conducting the experiments, the following hypotheses were proposed based on expected model behavior and findings from existing SER literature.

3.1 MFCC + SVM**Expected Outcome:**

- Expected to reach moderate accuracy (around 60–70%), but inferior to transformer-based systems.
- Should perform reasonably well as a classical baseline.
- Likely to struggle with subtle emotional differences.

Justification:

- MFCCs summarize the spectral envelope but lose detailed temporal information important for emotion expression.
- The merged corpus contains large speaker variability (age, gender, accent) which MFCC statistics are known to be sensitive to.
- Emotions such as Disgust and Fear have overlapping acoustic signatures, which MFCCs struggle to separate.

- However, strong emotions with distinctive spectral cues (Angry, Sad, Surprised) should be detected reliably.

3.2 ZERO SHOT TRANSFORMER

Expected Outcome:

- Expected to produce low accuracy (below 25%).
- Model trained on different corpora may not generalize to this mixed dataset.
- Predictions likely biased toward a few dominant emotions, especially Neutral.
- The model serves mainly as a benchmark showing the limits of zero-shot SER.

Justification:

- Pretraining corpora and emotion labels do not match the combined dataset used in this project.
- Emotional expression styles vary strongly between datasets:
 - RAVDESS: strongly acted
 - CREMA-D: crowd-rated naturalistic
 - SAVEE: British male actors
 - TESS: older female speakers
- Acoustic domain shift (microphones, background noise, speaking rate) greatly reduces zero-shot transferability.

3.3 FINE-TUNED Wav2Vec2

Expected Outcome:

- Expected to achieve the highest accuracy among all approaches (around 75–85%).
- Fine-tuning should help the model adapt to the combined dataset distribution.
- Should perform well across most emotion categories, including minority classes.
- Anticipated to produce more stable and balanced predictions than classical methods.

Justification:

- Wav2Vec2 benefits from self-supervised pretraining on thousands of hours of speech, learning:
 - robust phonetic and prosodic features
 - long-range temporal dependencies
 - speaker-invariant representations

- Fine-tuning adapts the pretrained embedding space to emotional distinctions in the merged dataset.
- Prior literature consistently reports strong gains from fine-tuning Wav2Vec2 for SER, even with limited labeled data.

4. FINDINGS AND RESULTS

The three models show clear differences in performance, reflecting their underlying modeling capabilities and sensitivity to cross-corpus variability.

4.1 MFCC + SVM

Overall Performance

- Test Accuracy: ≈ 0.72
- Macro F1-score: ≈ 0.74

Class	Precision	Recall	F1-Score	Support
Angry	0.78	0.78	0.78	325
Disgusted	0.56	0.59	0.57	280
Fearful	0.75	0.69	0.72	307
Happy	0.71	0.64	0.67	325
Neutral	0.72	0.76	0.74	269
Sad	0.72	0.79	0.75	326
Surprised	0.96	0.91	0.94	89
accuracy			0.72	1921
macro avg	0.74	0.74	0.74	1921
weighted avg	0.72	0.72	0.72	1921

Class-level Observations:

- Emotions like Happy, Angry, and Neutral showed relatively good recall due to distinct spectral signatures.
- Disgusted (recall ~ 0.59 , F1 ~ 0.57) and Fearful were frequently confused, reflecting overlapping spectral patterns.

- Fearful shows some confusion with other high-arousal emotions

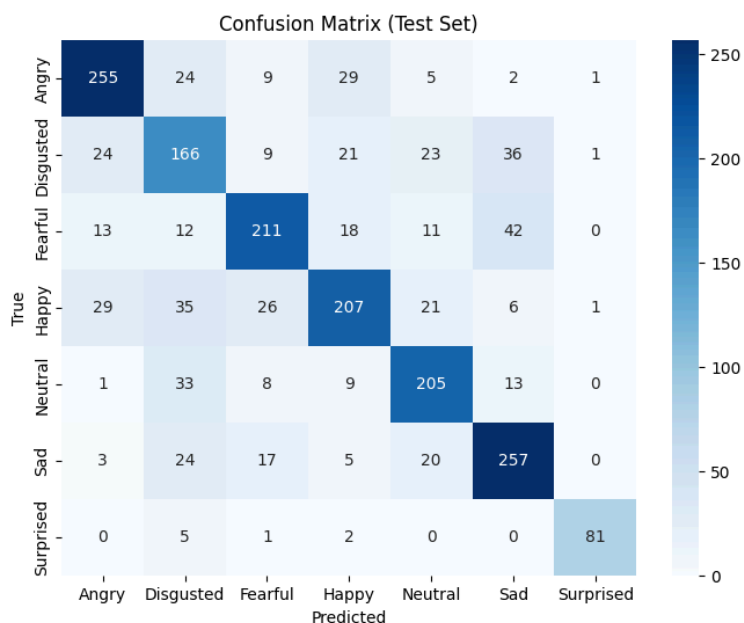
Confusion Matrix Interpretation:

- Misclassifications concentrated between Disgusted ↔ Fearful
- Some confusion between Happy ↔ Angry, likely due to similar high-energy features
- Surprised was extremely well recognized ($F1 \approx 0.94$), possibly due to sharp onset and high-pitch bursts captured well by MFCCs

Interpretation:

The MFCC + SVM baseline performs surprisingly well considering its simplicity. It captures enough spectral information to distinguish broad emotional categories. However:

- It cannot fully exploit sequential patterns (prosody over long context)
- It is sensitive to cross-corpus variability (different microphones, acting styles)
- Emotions with overlapping acoustic patterns (Disgusted, Fearful) are harder to separate



4.2 ZERO-SHOT TRANSFORMER

Test Results (Zero-Shot):

- Test Accuracy: ≈ 0.15
- Macro F1-score: ≈ 0.13

Class	Precision	Recall	F1-Score	Support
Angry	0	0	0	325
Disgusted	0.05	0.04	0.05	280
Fearful	0.04	0.02	0.03	307
Happy	0.07	0.1	0.08	325
Neutral	0.17	0.54	0.25	269
Sad	0.51	0.25	0.33	326
Surprised	0.3	0.11	0.16	89
accuracy			0.15	1921
macro avg	0.16	0.15	0.13	1921
weighted avg	0.15	0.15	0.12	1921

Class-level Observations:

- Strong bias toward Neutral (recall = 0.54)
- Most other emotions received very low recall (≈ 0 –10%)
- Predictions do not align with training distribution of our dataset

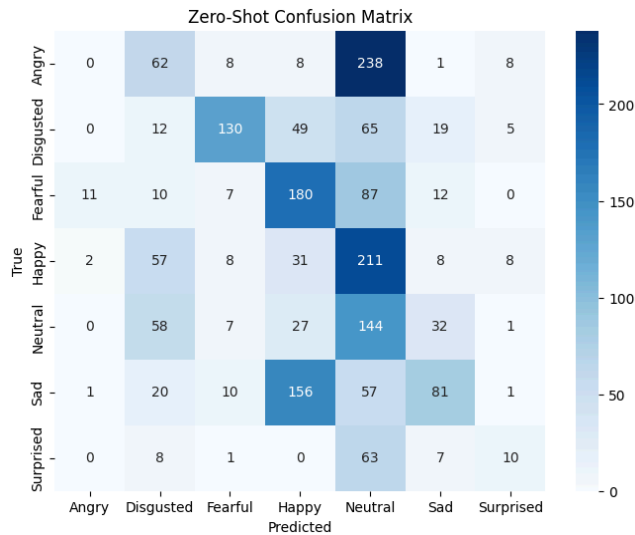
Confusion Matrix Interpretation:

- The model collapses most predictions into Neutral
- Happy and Sad receive some predictions but very inaccurately
- Minority class Surprised is nearly never predicted

Interpretation:

The zero-shot model performs poorly on this merged dataset. This confirms that emotion recognition is not easily transferable across datasets without adaptation:

- Emotional label sets differ
- Actors, languages, speaking styles, and recording conditions differ
- The model's decision boundaries do not match the new domain



4.3 FINE-TUNED Wav2Vec2

Test Results (Fine-Tuned):

- Test Accuracy: ≈ 0.79
- Macro F1-score: ≈ 0.80

Class	Precision	Recall	F1-Score	Support
Angry	0.78	0.9	0.84	325
Disgusted	0.85	0.67	0.75	280
Fearful	0.78	0.68	0.73	307
Happy	0.82	0.73	0.77	325
Neutral	0.77	0.92	0.84	269
Sad	0.72	0.8	0.76	326
Surprised	0.95	0.91	0.93	89
accuracy			0.79	1921
macro avg	0.81	0.8	0.8	1921
weighted avg	0.79	0.79	0.79	1921

Class-level Observations:

- High precision and recall across all emotions

- Angry, Neutral, Sad, Surprised all above 0.80 F1
- Disgusted remains slightly challenging (recall 0.67), but improved compared to SVM

Confusion Matrix Interpretation:

- Angry shows high recall (0.90), reflecting robust modeling of high-arousal cues
- Neutral shows the most improvement vs SVM, reaching recall 0.92
- The confusion matrix is much more diagonal than both the MFCC and zero-shot cases, with fewer misclassifications across emotion categories.

Training Curve Analysis:

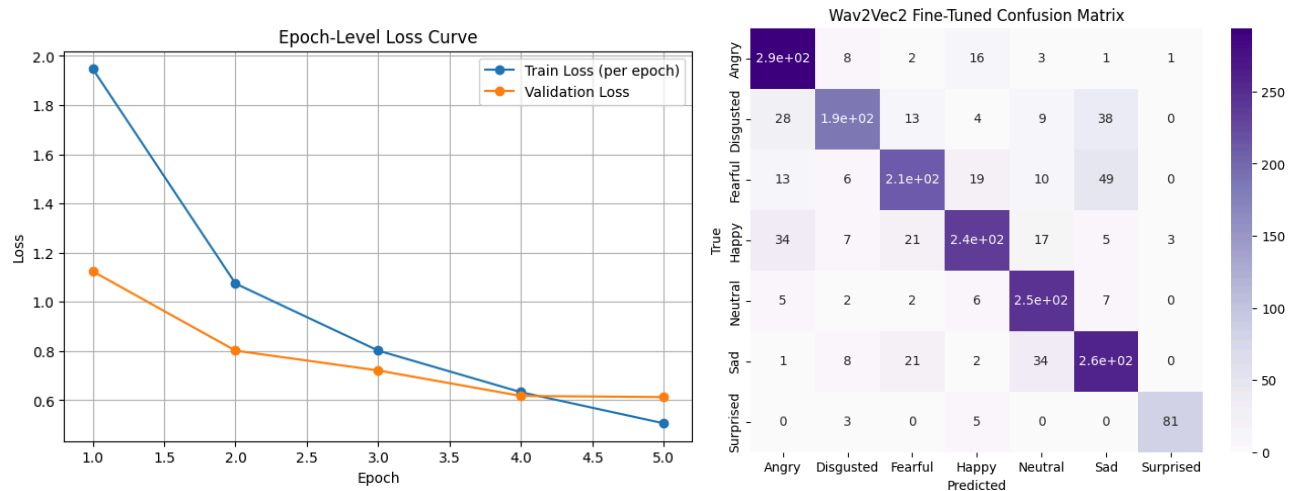
- Smooth decline in training loss
- Consistent reduction in validation loss through epochs
- No signs of overfitting; curves remain convergent

Interpretation:

Fine-tuning Wav2Vec2 on the combined corpus yields the best overall performance. This matches what has been reported in recent SER work using self-supervised transformers.

The model:

- Learns emotionally relevant patterns directly from raw audio
- Is robust to speaker and corpus variability
- Handles moderate class imbalance without special tricks



5. COMPARISON OF FRAMEWORKS

5.1 METRIC COMPARISON

Model	Accuracy	Macro F1
MFCC + SVM	0.7194	0.74
Zero-Shot Transformer	0.1483	0.13
Fine-Tuned Wav2Vec2	0.7902	0.8

5.2 ARCHITECTURAL AND COMPUTATIONAL COMPLEXITY

Model	Features	Parameters / Complexity	Training Required	Hardware
MFCC + SVM	MFCC stats	Low (SVM classifier)	Yes (fast)	CPU sufficient
Zero-Shot Model	Learned	Medium (pretrained transformer)	No training	CPU/GPU for inference
Fine-Tuned Wav2Vec2	Learned from waveform	High (Transformer backbone + head)	Yes (slowest)	GPU strongly recommended

Summary:

- MFCC + SVM is lightweight and good as a baseline.
- Zero-shot is computationally cheap (no training) but not usable in practice here.
- Fine-tuned Wav2Vec2 is computationally heavy but delivers the best accuracy and F1.

6. ADVANTAGES AND DISADVANTAGES OF EACH METHOD

6.1 MFCC + SVM

Advantages:

- Simple and fast to train
- Interpretable features
- Works reasonably well ($\approx 72\%$ accuracy)
- Good baseline for comparing more complex models

Drawbacks:

- Relies on handcrafted features; cannot capture all emotional nuance

- Limited modeling of temporal dynamics
- Performance degrades with cross-corpus variability

6.2 ZERO-SHOT TRANSFORMER

Advantages:

- No training required
- Easy to prototype and test
- Uses strong audio representations

Drawbacks:

- Very low accuracy on this dataset (~15%)
- Emotion label definitions differ from training corpora
- Poor generalization → not suitable for deployment without fine-tuning

6.3 FINE-TUNED Wav2Vec2

Advantages:

- Best performance ($\approx 79\%$ accuracy, macro F1 ≈ 0.80)
- Learns from raw waveforms, no manual feature design
- Robust across combined corpora and speakers
- Confusion matrix shows strong separation between emotions

Drawbacks:

- Requires GPU and more training time
- Larger model size and memory footprint
- More complex to set up and tune

7. CONCLUSION

This assignment evaluated three emotion recognition frameworks on a merged emotional speech dataset (RAVDESS, CREMA-D, TESS, SAVEE):

- The MFCC + SVM baseline reached $\approx 72\%$ accuracy, confirming that classical methods still provide a strong reference point.
- The Zero-Shot Transformer performed poorly ($\approx 15\%$ accuracy), showing that pretrained emotion models do not directly generalize to new corpora without adaptation.

- The Fine-Tuned Wav2Vec2 model achieved the best performance, with $\approx 79\%$ accuracy and macro $F1 \approx 0.80$, consistent with recent literature on self-supervised transformers for SER.

Overall, the experiments support the hypothesis that fine-tuned transformer-based models are currently the most effective approach for Audio Emotion Recognition, especially in challenging cross-corpus settings. The MFCC baseline and zero-shot results provide useful context and demonstrate the clear benefit of end-to-end fine-tuning.

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